

Introduction

Blood shortages have become a prevalent issue all over the world, with blood not being available for transfusions due to issues with type matching and availability, since only **3%** of the population donates blood.

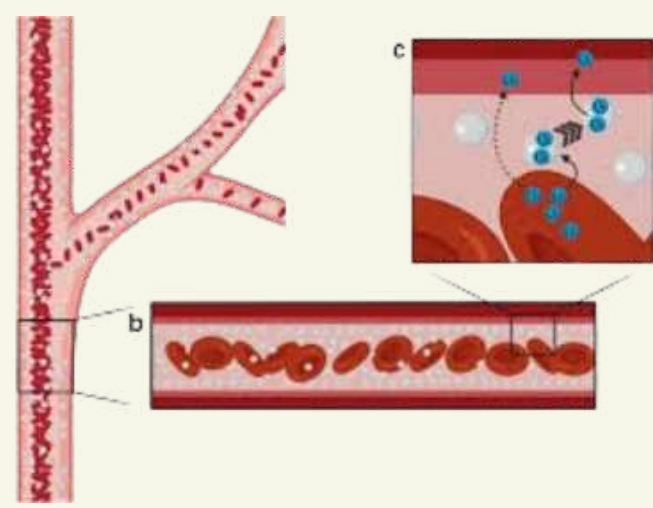


Image from Jagers et al. (2020)

This makes it viable for use as an **artificial blood oxygen therapeutic** to increase oxygen content in blood. **Ringer's Solution** is an IV fluid composed of water and numerous salts, and is often used to restore body fluids after an injury.

Research Question

How does a 35% Ringer's Solution + 62% PFOB + 3% lecithin mixture mimicking blood affect lactate concentration and physical conditions of *Lumbricus terrestris* compared to PFOB or Ringer's Solution alone under hypoxic conditions?

Hypothesis & Variables

Hypothesis:

Ringer's Solution increases fluid volume and nutrients while PFOB dissolves and transports oxygen efficiently but can impair bodily functions when administered alone. Therefore, earthworms treated with the artificial blood solution will exhibit the lowest lactate concentrations and healthiest outward physical characteristics.

IV: Treatment Group:

- 1) No Treatment (Distilled Water)
- 2) Ringer's Solution Only
- 3) Emulsified PFOB only
- 4) 35% Ringer's Solution + 62% PFOB + 3% Powdered Soy Lecithin Mixture

DV: L-Lactate Concentration of earthworm hemolymph samples

Control: Earthworms treated with water only

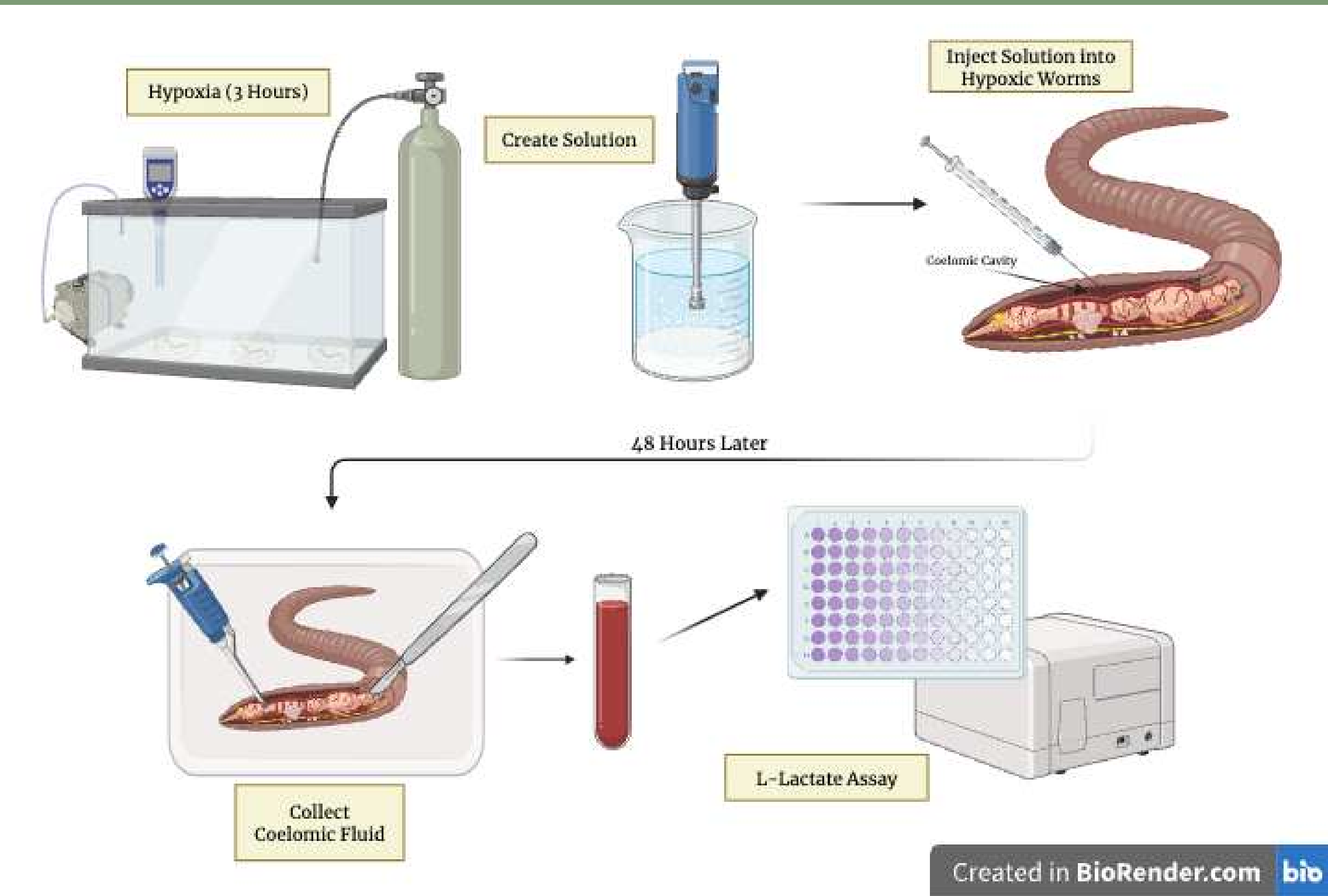
The Effects of a PFOB and Ringer's Solution Emulsion as an Artificial Blood Substitute in *Lumbricus terrestris* via Coelomic Injection

Methods

Materials:

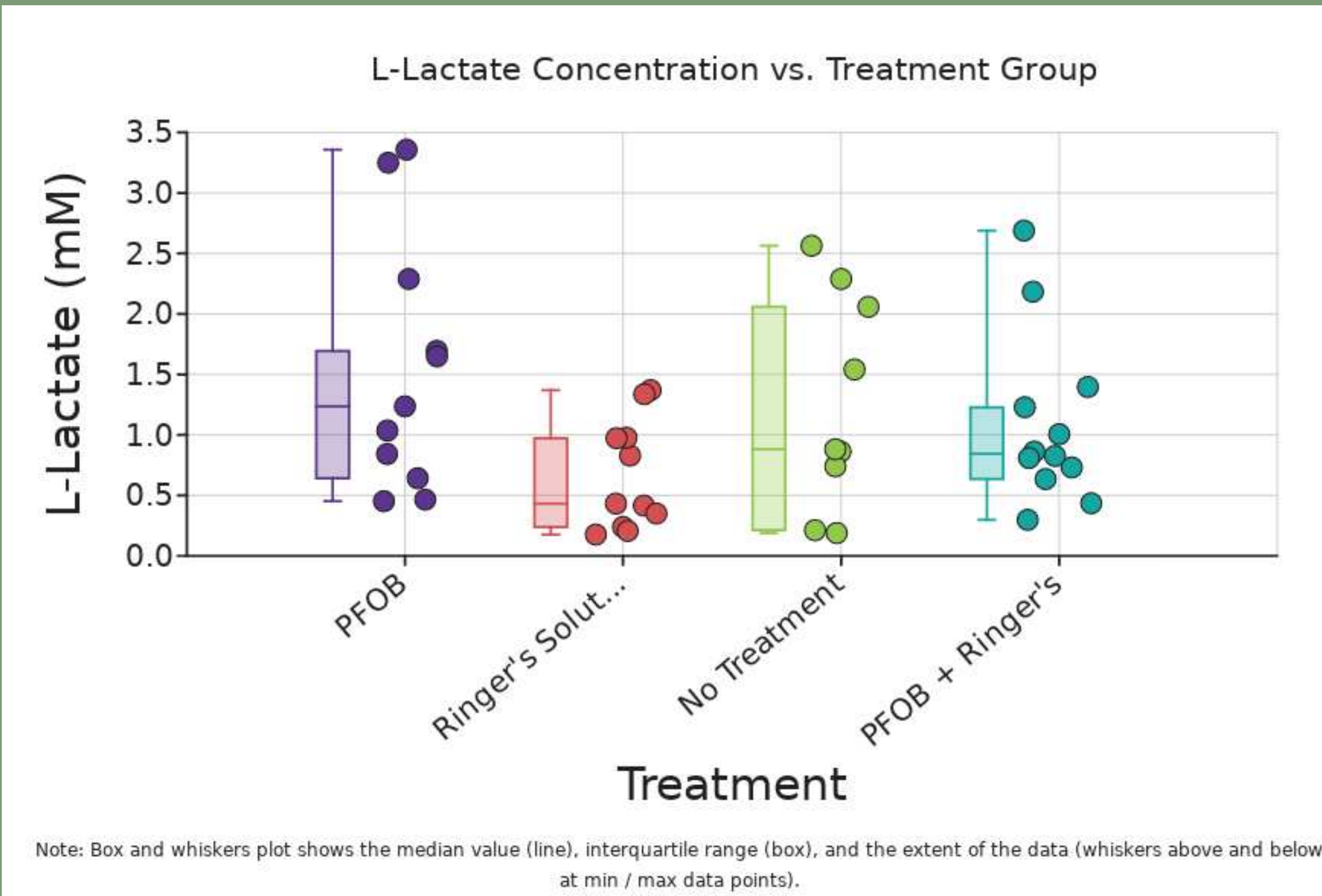
- Perfluorooctyl Bromide
- Ringer's Solution
- Powdered Soy Lecithin
- Oxygen Meter
- *Lumbricus terrestris*
- Plate Reader
- Spectrophotometer
- L-Lactate Assay Kit
- Nitrogen Gas Cylinder
- Air Pump
- Insulin syringe

1. Put worms in 10% oxygen hypoxia chamber for 3 hours.
2. Create treatment solution
 - a. Dissolve Ringer's Solution tablet
 - b. Emulsify PFOB with 5% powdered soy lecithin
3. Inject solution into hypoxic worms with insulin syringe.



- 4) After 48 hours, use dissection tools to cut through earthworm skin. Collect internal coelomic fluid using micropipette.
- 5) Perform L-Lactate Assay on collected hemolymph samples.

Results



ANOVA TEST	
p-value	0.09

An **ANOVA test** was run in DataClassroom, resulting in a p-value of **0.09**, which, with a significance level of 0.10, indicates strong evidence of a statistically significant difference in lactate concentrations between treatment groups.

Post-Hoc Tukey's Range Test	
Pair:	p-value
Ringer's Solution - PFOB	0.06
No Treatment - PFOB	0.86
PFOB + Ringer's - PFOB	0.54
No Treatment - Ringer's Solution	0.35
PFOB + Ringer's - Ringer's Solution	0.57
PFOB + Ringer's - No Treatment	0.96

A **post-hoc Tukey's Range test** was conducted to analyze if there is significant differences in lactate concentration between treatment groups. There seems to be strong indication that the Ringer's Solution only and PFOB groups had significant differences in lactate concentrations, and weak indication that there is some difference between the No Treatment and Ringer's Solution only group.

Analysis

- Since the **p-value** found through statistical tests was 0.09, greater than the significance level of 0.10, the tests indicate there is strong evidence of significant differences in lactate concentration between the treatment groups.
- However, slight trends are visible, particularly between Ringer's Solution and PFOB groups, which had a p-value of 0.06, indicating that **PFOB administered on its own may increase lactate concentrations** when compared to Ringer's Solution administered on its own.
- Post-hoc Tukey's Range Test showed high p-values across pairwise comparison; no statistically meaningful differences.

Conclusion

The results of this experiment do not support the original hypothesis that a combination of PFOB and Ringer's Solution will increase oxygen levels and therefore decrease lactate concentrations as the p-value of 0.96 indicates no evidence of a significant difference between groups, but the ANOVA p-value of 0.09 indicates that the substances may have some impact on lactate concentration, supporting the idea that PFOB and Ringer's Solution have potential to be used as a blood oxygen therapeutic.

Limitations to this research include a lack of trials, inconsistency in procedure using the earthworms, control of hypoxic conditions, time restraints, etc.

Future Research

- Future research should test this solution under more precise hypoxic conditions and consistent injection and dissection techniques.
- Other organisms, particularly vertebrate organisms with closed circulatory systems, should be tested with the same procedure to more accurately emulate the human circulatory system.

References

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